Arthritis and Post-Infection Syndromes in Lyme Disease

Arthritis in patients with Lyme disease typically responds to immunosuppressive agents such as methotrexate and antitumor necrosis factor therapies, leading to suggestions that these patients may have an autoimmune syndrome. Molecular mimicry between borrelial antigens and host proteins has been proposed as one possible mechanism, but this remains unproven.

There have been numerous reports of patients developing a fibromyalgia-like syndrome following *B. burgdorferi* infection. This syndrome includes profound fatigue, depression, myalgias, polyarthralgias without arthritis, paresthesias, and neurocognitive difficulties such as memory impairment, concentration deficits (particularly auditory), and decreased mental flexibility (e.g., impaired verbal fluency). This condition is often referred to as chronic Lyme disease. However, the association between these symptoms and *B. burgdorferi* infection remains controversial. Several large epidemiologic studies have shown that individuals with a history of Lyme disease are no more likely to experience these symptoms than the general population.

Diagnosis of Lyme Disease

Clinical evaluation based on physical findings, history of tick exposure, known tick bites, or symptoms consistent with the multi-system presentation of Lyme disease can aid in diagnosis. In cases of classic erythema migrans (EM) or Bannwarth syndrome, the diagnosis can be made clinically without laboratory confirmation. Diagnostic challenges arise when patients present with skin, cardiac, neurologic, or musculoskeletal symptoms without a history of EM or tick exposure.

Laboratory Tests

Direct Detection of *B. burgdorferi*

Culture of *B. burgdorferi* is considered the gold standard for diagnosis. However, it is impractical in most clinical settings due to the need for specialized media and the organism's slow growth. *B. burgdorferi* can be successfully cultured from skin biopsies of EM and acrodermatitis chronica atrophicans (ACA), as well as from cerebrospinal fluid (CSF) in patients with neuroborreliosis. Recent studies indicate that the organism may be isolated from blood in up to 50% of untreated adults with EM. However, culture sensitivity is low in patients with extracutaneous manifestations, especially in later stages of disease.

Polymerase chain reaction (PCR) assays are highly sensitive for detecting *B. burgdorferi* DNA in skin biopsy specimens from EM lesions and in synovial fluid from patients with Lyme arthritis. However, PCR testing for *B. burgdorferi* DNA has not been approved by the U.S. Food and Drug Administration (FDA), and its routine clinical use is not recommended.

Serologic Diagnosis of Lyme Disease

Immunologic testing is the primary laboratory method used to support a clinical diagnosis of Lyme disease. In the United States, the FDA has approved over 70 different immunoassays—primarily enzyme-linked immunosorbent assays (ELISA) and Western immunoblot assays.

ELISA tests typically use whole-cell lysates of *B. burgdorferi* sensu lato or, less commonly, purified antigens to detect anti-*B. burgdorferi* IgM and/or IgG antibodies. While relatively specific, ELISA tests have a high rate of false-positive results. Conditions associated with false-positive IgM include autoimmune diseases (e.g., lupus, rheumatoid arthritis), Epstein-Barr virus infection, bacterial endocarditis, and other tick-borne diseases. False-positive IgG results may occur in syphilis, *Helicobacter pylori* infection, systemic lupus erythematosus, ehrlichiosis, and babesiosis.

A newer ELISA that targets a specific peptide from the VlsE protein of *B. burgdorferi*—known as the C6 peptide test—demonstrates improved specificity compared to traditional whole-cell ELISA. This test detects only IgG antibodies, but IgG to the C6 peptide appears early in infection. Its sensitivity in early Lyme disease is comparable to that of IgM-based whole-cell ELISA tests.